Stolen Strata Development Log

> **How to read this log.** It is a working diary, written as the work happened. Entries 1–15 describe the first version of the project (saffron, roads, settlements, "25 degraded terraces", 1.84% → 8.43%). Those results were withdrawn in October 2026: they came from comparing two differently built satellite products (Entry 18). Entries 16 onward describe the rebuild. The research questions in Entry 1 (RQ1–RQ4) were replaced by the three questions in Section 1 of `SS_Research_Paper.md`. For the current findings read the paper or the executive summary, not this log.

Am studying Karewas in Kashmir. These are high, flat topped mud plateaus from a very old dried up lake, nd this special soil is the exact thing saffron plants need to grow properly. For many years, people have been illegally digging up this mud and building random houses on top of them (but nobody evr calculated the destruction with proper numbers before this). So, I built a fully automated code pipeline from scratch, using DEM elevation data all the way to a final web dashboard, to trace out every single plateau border using math, track how they became plain empty mud over a 31-year satellite timeline, nd link this physical destruction to the saffron economy sitting next to it (as value at risk, not realised loss), the new roads tht help trucks carry mud away easily, nd the total lack of strict government rules.

My log covers everything. I built a terrain analysis code using TPI nd slope cutoffs to replace lazy hand drawn maps, nd I also wrote down my bad starts with the multi year NDVI code which I had to fix before the main destruction statistics made any sense. (Plus, I added notes from my multiple revision rounds where I pushed my math checks and limitations way further than my easy 1st draft did).

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Entry 1

Am looking at Kashmir Karewas. They are basically high, flat topped mud hills tht got left behind when a massive old lake in the main valley floor dried up over the Plio-Pleistocene (hundreds of thousands to millions of years ago). I found tht these shapes are not jst old rocks, because their top soil is made of loose, well drained loess mud which is the exact and only reason why saffron plants (Crocus sativus) grow so well here. This saffron is super expensive and has a GI tag, so thousands of farming homes in the Pampore region fully depend on it to eat nd live. If you check the geology, these Karewa sediments filled up the ground because the Pir Panjal mountains pushed up nd blocked the water flow long ago. Old papers divide the mud into two main layers: a Lower Karewa layer made of old lake clays n dirty lignite coal from when deep water was standing still, nd an Upper Karewa layer with rough river gravels nd sand from when the lake was slowly choking n filling up. All this mud got stuck during the Plio Pleistocene era. Later, the Jhelum river cut deep lines through the valley, washing away most of the mud but leaving behind these separate flat hills (nd wind blown dust from the ice age left a final silt layer on top). This calcareous silt soil is perfect for saffron bulbs. My project focuses on this exact core point: how a weird old lake shape controls a whole modern state economy.

The terraces are disappearing fast. Over the last thirty years, people hve been illegally digging out the mud to make bricks n roads, plus cities are growing wildly, so these beautiful flat hills are getting flattened completely. Local newspapers talk abt this damage sometimes, but I found zero proper satellite data tracking it over the decades (n nobody connected the physical land destruction to the loss of the saffron business).

I spotted three massive blanks in the old studies. 1st, nobody ever calculated the total area of Karewa loss using multi decade satellite records. Only random stories n tiny local reports existed. 2nd, no map existed tht shows where the hills were broken right next to where saffron grows, so I did not know how much farming soil was destroyed. 3rd, I wanted to check if big government projects like the National Saffron Mission are spending money on safe, solid land or on spots tht are already getting ruined by miners.

So, I wrote down four research questions. My RQ1 checks the total area change of Karewas from the oldest Landsat satellite records up to now. My RQ2 asks where the damage is happening most n if it sits close to highways, town centers, or brick kilns. My RQ3 asks how much destroyed hill area used to be saffron farms n wht that means for the farmers' income. Lastly, my RQ4 checks if government money is going to the right spots where the land is still safe nd whole.

Have clear tasks set up now. 1st, I need to read and summarize all the old geology papers on how Karewas were formed, their soil layers, and their types so I have a strong theory base. 2nd, I will use DEM elevation models to trace out the exact flat plateau borders using automated math checks n then double check them against regular satellite photos. 3rd, I will build a long term land cover tracking code using historical Landsat n

Sentinel-2 archives to calculate exactly how much Karewa soil was lost to open mining, brick kilns, and urban housing construction over the longest possible years. 4th, I will map the boundaries of saffron farms inside the Pampore region nd lay that timeline right on top of my land destruction maps.

Then I will look at this as a serious political and planning issue. Am going to check if government money nd agricultural policies are matching up with the areas where the land is still safe nd unbroken on the ground. Finally, I will put all my hard work out for the public by launching an open code repository, an online dashboard, a proper research paper draft, n detailed GIS maps that look as good as the rest of my portfolio.

My main focus area is central Kashmir. I selected the major Karewa spots across Pampore, Pulwama, n Budgam districts, along with the classic old Zewan section near Srinagar (this specific zone has the maximum number of saffron growing flat hills nd every book calls it a total hotspot for illegal soil theft). My bounding box coordinates sit between 33.85°N to 34.15°N nd 74.75°E to 75.15°E. To do my long term land cover work, I chose to download Landsat 5, 7, 8, and 9 data archives via Google Earth Engine, n I also pulled sharp Sentinel-2 images for checking recent years.

I listed more datasets to look for. For the flat plateau boundary mapping, I planned to use Copernicus DEM nd Cartosat DEM data from the European Space Agency nd ISRO's Bhuvan portal to catch the sharp slope breaks, plus old paper topographic maps. (For my economics and policy checks, I planned to get official saffron farm boundary maps from the J&K Department of Agriculture, plus National Saffron Mission booklets). [Correction, Oct 2026: in the end only the Copernicus DEM was used. No Cartosat DEM, topographic maps, saffron boundary maps or NSM spatial data went into the analysis, see Entries 4, 11 and 12.]

I am following four clear steps for my code flow. 1st, I am using DEM data to map out the flat hill tops n trace their boundaries. 2nd, I am setting up a multi year land classification script to separate the untouched Karewa surfaces from the mined mud pits, new buildings, nd farming land across all the available satellite collections. 3rd, I will run a change detection code to measure the total destruction speed n pinpoint exactly which villages are becoming the biggest mining hotspots. (Finally, I will lay the saffron crop boundary layers directly over the broken land maps nd look at the government spending data, making sure I clearly write down the missing land lease files that the local department does not show online). For my final submission, I am handing over a fully working public GitHub code folder, an online interactive chart dashboard, a proper research draft, this exact project dev log, nd clean high resolution static map figures.

This whole project gives solid satellite proof to a major farming nd land crisis tht people only talked abt in news stories before. So I am directly combining old school geology ideas (like how lake layers form nd how hills get cut by rivers) with new remote sensing tools n GIS mapping layers. Then, I am using this data to check a major planning question abt whether agriculture funding is going to the wrong spots on the map. That's proof I can handle both physical geography n human social policies nicely for this portfolio.

Entry 2

So I am starting my 1st coding task. My main plan is to write a clean, automatic script pipeline that can find and trace out the exact flat Karewa terrace edges across Kashmir without relying on hand drawing tools. (Am using a code-1st approach here. So QGIS will only be used at the very end just to look at the final map images nd check if they look correct).

1st, I built my project folders. So I set up folders named `data/`, `src/`, `notebooks/`, `outputs/`, `dashboard/`, `docs/`, nd `tests/` so tht my raw satellite files, Python scripts, intermediate steps, n final report charts stay completely separate n clean. Nxt, I opened the Google Earth Engine platform online, typed in my coordinates

for the Pampore, Pulwama, Budgam, and Zewan regions, and exported the 30-meter Copernicus GLO-30 DEM elevation raster file. I downloaded this raw terrain file and saved it directly inside my `data/raw/` folder so my scripts can read it.

I picked a specific math path. So I chose to use the Topographic Position Index (TPI, Weiss 2001) mixed with a simple slope angle cutoff so my script can automatically find ground tht is both totally flat on top n higher than the surrounding valleys. This is the exact shape signature of a true Karewa terrace. So I chose this code approach on purpose so I do not hve to sit and hand draw hundreds of borders inside QGIS (which would be super boring nd impossible to repeat exactly later).

But I ran into a massive wall of code bugs. 1st, the separate slope map I downloaded from Earth Engine came out completely blank nd corrupted, so I fixed it by writing a local Python function to calculate the slope math directly from my raw DEM file. Then, I realized the original DEM was using regular geographic degrees, which makes it totally impossible to calculate meters for slope. Fixed this by reprojecting the whole image map into the UTM Zone 43N coordinate grid system. (This reprojection step accidentally created a thin, messy edge of dead pixels around the image boundaries which would hve ruined my TPI smoothing filters, so I trimmed those edges away and used a nearest neighbor fill to clear the remaining gaps).

Next, a stupid code check told me tht my whole map array was empty n broken! This was a total false alarm because NumPy's standard min nd max formulas break completely n return NaN if even a single bad pixel exists in ur data grid. I solved this scare by switching to NaN aware formulas like `np.nanmin` nd printing explicit numbers, which proved tht 99.6% of my raster grid was perfectly fine. Lastly, my final binary mask file (which uses uint8 format) could not save the original DEM's nodata value number, so I manually coded the mask's nodata setting to `None` inside my script.

Tested my numbers 1st. In my starting run, I set the code to look for a TPI greater than 5 meters n a slope under 5 degrees, which threw out 3,053 messy raw shapes, but after I cleaned it up to keep only shapes bigger than 0.05 square kilometers, I was left with jst 115 pieces. When I opened the map to look at them, I saw these shapes were very weird, skinny, n long like threads instead of big wide blocks. (This meant my settings were way too strict n were catching jst the very top ridges of the hills instead of the full flat tops). So I changed my values to be more relaxed: TPI greater than 3 meters and slope under 8 degrees, which gave me 6,789 raw shapes, and keeping the same size filter left me with 201 proper candidates. Next, I added an extra elevation filter to keep only things between 1,550 nd 2,000 meters because that is where Karewas sit next to the valley floor, but it did not delete a single shape, telling me my filters match each other perfectly.

The check looked good. I put my final 201 shape boundaries on top of standard Google satellite map images, nd I could clearly see them tracing the exact high ground tht sits away from the low, green forests. Even better, my shapes sat right on top of the map txt tht says "Saffron Fields, Lethpora" near Pampore! This is jst a solid visual check tht the shapes look like true Karewa terrain, not formal ground-truth validation. Tht would need properly labelled reference points, which I don't have yet at this stage. 😀

So, my 2nd big milestone is done. I have saved all 201 filtered terrace shapes inside `data/interim/karewa_candidates_filtered.gpkg`, n I will use these exact map pieces for my nxt big task: writing the long term code to track how much of this land is getting destroyed by miners n new house builders.

Entry 3

> [Withdrawn in v2. The result described in this entry depended on the withdrawn first-version layer; see the note at the top and Entry 18.]

I started my nxt task. I wanted to see exactly how much the land got ruined inside my 201 approved Karewa boundaries by checking plant greenness between an old 1994 baseline n our current 2025 period, using satellite NDVI scores to separate green farms from brown mud mining pits nd new concrete structures. Fetched my satellite pictures using Earth Engine code. Specifically, I grabbed a bunch of old Landsat 5 images from 1993 to 1996 to map the past, pulled sharp Sentinel-2 pictures from 2024 to 2025 for the modern layer, calculated the NDVI greenness formula for both maps, nd ran zonal stats inside all 201 hill shapes to see the difference over time.

The 1st code output was totally backwards (nd tht caught me completely off guard). My data showed tht the average NDVI score went up over these thirty years. Ugh. This made no sense at all because open cast mud mining is supposed to kill the green fields, so instead of blindly trusting this weird upward graph, I immediately stopped everything n treated it as a huge methodology error that I had to investigate nd fix right away.

Found two big blunders. 1st, I noticed a huge seasonal mismatch because my old 1994 data stack used satellite pictures from the whole year (including freezing winters when crops are completely dead nd brown), while my 2025 data only looked at the lush summer months. This error made the two maps completely useless for comparisons since the summer pictures would always look way greener even if miners dug up half the hill! I fixed this by cutting down my 1994 Landsat timeline to the exact same June to September monsoon months that I used for the 2025 Sentinel-2 dataset.

2nd, my math metric was totally blind. Taking the single average NDVI score for an entire big hill shape washed out the signal from tiny, deep mining holes because the remaining huge green fields hid the damage. (So, a truck could steal tons of soil from a corner n my old code would not even notice a change). So I threw away the average NDVI idea nd created a new bare earth fraction metric instead. Now, my script counts the exact percentage of pixels inside each terrace shape tht drop below a low NDVI number to see how much bare ground appeared between the two times.

My new calculations worked perfectly. With my twin fixes running in the code, I found tht the total average bare mud share inside all 201 hills jumped from a tiny 1.8% back in 1994 up to a big 8.4% in 2025 (tht comes out to a massive 4.6 times increase in empty dirt). Out of my 201 total shapes, exactly 25 hills showed an extreme jump in their empty soil share by 15% or even more. I tagged these 25 bad spots as heavily ruined. The remaining 176 hills are thankfully still green nd keeping their original farming look.

So my 3rd milestone is officially checked. I saved this new dataset directly at `data/processed/karewa_bare_earth_change.gpkg`. Now, every single flat hill shape holds its old 1994 bare soil score, its new 2025 score, the exact change value between them, and my final ruined versus safe tag. This is the very 1st solid number proof of my project. Nxt up, I will double check these exact places on a visual map, and then I will drop my saffron crop boundary layers on top to see how badly this soil theft is hurting the local farmers' wallets.

Entry 4

> [Withdrawn in v2. The result described in this entry depended on the withdrawn first-version layer; see the note at the top and Entry 18.]

So I needed a visual check. Had to see if the 25 hills tagged as ruined by my code script were true, on-the-ground mud pits or jst some stupid error from my math code pipeline. So I opened QGIS n loaded the file `data/processed/karewa_bare_earth_change.gpkg` right over a live Google satellite base map, changed the styling colors based on the status column to show ruined versus safe areas clearly, nd scrolled around with my mouse to check the locations by eye.

The ruined spots were not random. They all clustered together nicely. Found the biggest group of broken hills right near Pari Gam Khalsa n Newa Pulwama, sitting perfectly on top of a massive, tan colored bare patch of ground on the satellite image which looks exactly like a working mining quarry or giant dirt pit compared to the green farms next to it. (Then I saw another smaller ruined shape sitting right nxt to the map marker for "Saffron Fields, Lethpora," which perfectly links this land theft to the main saffron capital area that my paper talks abt!). Also spotted a couple of lonely ruined hills sitting far away near the Srinagar airport n around Wuyan, but they were jst single broken spots instead of a big group.

The layout looks very solid. My mapping pattern matches true, local land destruction instead of random computer code glitches, giving me great visual backup for my bare earth fraction numbers. This pinpointed a main zone (the Pari Gam Khalsa group) where I need to zoom in n check the satellite photos much closer. (It also proves that one broken hill sits right at the edge of a big known saffron farm area).

Later, I ran a sensitivity test using different distance buffers to be double sure. My calculation results show the affected share climbing up very smoothly across the test limits, going from 21% to 29%, then 43%, 71%, 86%, n finally 93% without any weird or crazy spikes anywhere. So my main 43%-within-1-km number isn't jst a random fluke from an arbitrary distance choice. It's a true n steady geographic fact. For my nxt step, I am going to find or digitize a proper saffron farming boundary layer for the Pampore region nd stack it right on top of my broken terrace maps so I can calculate the exact amount of farming soil lost to these mines.

Entry 5

> [Withdrawn in v2. The result described in this entry depended on the withdrawn first-version layer; see the note at the top and Entry 18.]

I changed my target goal. Needed to spot the saffron growing hills using their weird upside down lifecycle (they sleep in hot summers n grow green leaves only after winter flower season) so I can measure how close they sit to the already ruined mining hills.

But my 1st try broke down completely. So I calculated a new saffron index by subtracting summer NDVI from autumn NDVI, using an October to November green timeline, but my script threw out completely negative numbers across all 201 hill shapes. (This was a huge shock because it was the exact opposite of what the biology books say!). So I tracked this error down to crop timing. See, October n November is jst the flower picking time with almost zero green leaves on the ground, while the thick leaf canopy only grows out much later after the flowers drop during winter. Fixed this code blunder by shifting my green comparison timeline all the way to March because tht is when the post flowering plant leaves grow thick right after the winter snow melts away.

The new calculations worked. With the fixed March window running, I found tht exactly 14 out of my 201 hills are definitely growing saffron crops. None of these 14 pieces touch the 25 ruined mining spots directly, but my distance code showed that the closest saffron field sits a tiny 80 meters away from an active mud digging zone, nd 6 out of those 14 hills (tht is 43% of them!) sit within jst 1 kilometer of a mine. (I also ran a sensitivity test checking distances from 500 meters to 2.5 kilometers, nd the numbers climbed up very smoothly nd

steadily, proving this is a solid geographic pattern n not some lucky accident from a single chosen cutoff number).

So I found my main point. The overlay shows tht no saffron terrace touches a degraded terrace yet. Still, a huge portion of these premium farms sit dangerously close to expanding bare dirt pits. So my project is showing an encroachment risk. It is not a direct loss story. This is a safer claim than an overlap number.

Entry 6

> [Withdrawn in v2. The result described in this entry depended on the withdrawn first-version layer; see the note at the top and Entry 18.]

I needed a quick reality check. Had to test if my 14 saffron hill shapes made any sense, so I compared their total size against an official, separate report on how much saffron land exists in Pampore.

So I hunted down the proper paperwork. I found the FAO GIAHS report titled "Saffron Heritage Site of Kashmir in India" (Part-1, which SKUAST K helped write for the pilot project) nd this document clearly states that Pampore has over 3,200 hectares of saffron farming land supporting more than 17,000 farming homes. Then, I used my script to calculate the exact area of my 14 mapped saffron shapes to see how they match up.

My code output stopped me in my tracks. The total area of my 14 detected saffron hills came out to jst 225.4 hectares, meaning my code only caught abt 7% of that official 3,200-hectare FAO baseline. (Tht leaves a giant 93% data gap tht I need to explain right now).

I refused to believe this crazy gap. A giant drop like tht would mean saffron farming in Kashmir is almost 100% dead, which is totally wrong because independent news channels keep reporting tht saffron production recently hit a massive 25-year high. So, I figured out tht this mistake is jst a detection recall issue in my code. It isn't farming destruction happening in the world. My saffron leaf filter only scans inside those 201 hill shapes, which I already cut down heavily using my strict starting rules (like TPI, slope cuts, nd minimum size filters). (Stacking another strict winter leaf growth rule on top means my code only picks up a tiny, super sure group of pixels instead of counting every single farm in the valley).

This gap is a limitation of my method, not a fact on the ground, meaning my 14 shapes are jst a small, high accuracy group of saffron hills, not the whole map footprint. But this error does not break my distance risk finding (where 43% of my found saffron hills sit within 1 kilometer of a mine) because that number is just a relative spatial comparison inside my small group nd does not care if I missed other fields. I am still keeping tht official FAO GIAHS data (3,200 hectares and 17,000 farming homes) as a verified reference text for my paper introduction.

Entry 7

Needed a major headline number. Chose to change my relative percentage scores into plain hectare figures so I can show exactly how much physical Karewa land got turned into raw empty dirt between 1994 and 2025.

Did some quick multiplication inside my script. For every single one of my 201 hill shapes, I multiplied the total polygon area by its own bare earth percentage for both timelines, which gave me the exact count of bare hectares per hill, n then I added up all these values for both years separately. The gap between these two big totals gives me the net destroyed area. (Also ran this exact same math a 2nd time but locked it down only to the 25 heavily ruined hills to see if the soil theft is happening in jst one big cluster or spread out everywhere).

Got my final numbers. The total size of all my 201 mapped hill shapes added up to 3,305.3 hectares, which sits surprisingly close to tht official 3,200-hectare FAO saffron baseline (even though they track slightly different things like total hill space versus crop space, it is a very good sanity check on my map scales). [Correction, Oct 2026: this is not a valid check. Terrace area across three districts and Pampore saffron crop area measure different things, so their closeness is a coincidence, not a check on map scale.] My data shows bare mud area crawled up from just 32.2 hectares back in 1994 to a massive 222.6 hectares in 2025! Tht adds up to a net loss of exactly 190.3 hectares, or 5.8% of my total mapped terrace lands. (And here is the craziest part: abt 128.2 hectares, which is a massive 67% of the total destruction, happened inside jst those 25 hills I flagged as ruined, even though those bad hills make up only 9.7% of the total terrace map space).

This is my main project result. Between 1994 n 2025, exactly 190.3 hectares of premium Karewa hills turned into plain empty dirt, nd this damage is grouped heavily in specific spots instead of being scattered loosely everywhere because two thirds of the loss sits in a tiny 12.4% subset of hills. Combining this with my older risk calculation (where 43% of saffron fields sit within 1 kilometer of a mine) finishes my core statistics work. Now I can move away from heavy math nd start building my public facing stuff like maps, the online dashboard, n my final research paper draft.

Entry 8

> [Withdrawn in v2. The result described in this entry depended on the withdrawn first-version layer; see the note at the top and Entry 18.]

So I needed a better timeline. My old two point check from 1994 straight to 2025 was way too simple, so I chose to insert two middle checkpoints (2005 and 2015) to see if this Karewa land destruction is a slow, steady change over thirty years or if it jst boomed during one single bad decade. (This matches my main goal of using the absolute longest data chain instead of a quick shortcut).

So I jumped back into Google Earth Engine. I downloaded season matched imagery from the May to October window for both 2005 using Landsat 5 n 2015 using Landsat 8, ran the exact same bare earth percentage logic I built for my 1994 n 2025 steps, n calculated the final average empty dirt share across all 201 hill shapes for every single one of these four selected years.

My 2005 script crashed completely. My 1st Landsat 5 search matching that exact June to September monsoon block n low cloud limit gave me zero images for Kashmir, which completely broke my download code with a missing band script error. (I printed out the total image counts to check wht was happening nd realized the database folder was literally empty under those strict rules!). Solved this annoying code failure by stretching my months out from May to October, expanding the year search range to a wide nine year span from 2001 to 2009 centered right on 2005, n removing the cloud filter entirely. Instead, I let my script use a median composite formula across this huge pile of pictures to wipe out the cloud noise automatically.

I got my final four point trend. Calculated the average bare earth scores across all 201 hills over time: 1.84% in 1994, then 2.62% in 2005, then 2.63% in 2015, nd finally a huge 8.43% in 2025. The numbers for 2005 nd 2015 are almost identical, showing tht the hills stayed pretty calm nd unchanged during tht ten year block. (But then look at the massive jump up to 8.43% by 2025!). What tht tells me is almost all the land destruction happened very recently in jst this last decade alone; nd the damage speed right now is way bigger than the previous twenty years combined.

This fixes my old limitation. So I cleared up the two point date problem nd found a massive new point for my final project results: the mud hills are not getting ruined at a steady speed over thirty years. Instead, it is a

recent, super fast boom happening almost entirely between 2015 nd 2025. My saved dataset file `data/processed/karewa_multitemporal_trend.gpkg` now holds the bare mud percentage data for all four checked years inside every hill shape. (For my nxt step, I am going to run a proper math test to check if the broken hills sit systematically closer to highways or town centers compared to the safe hills to answer my 2nd research question using OpenStreetMap road network data files).

Entry 9

> [Withdrawn in v2. The result described in this entry depended on the withdrawn first-version layer; see the note at the top and Entry 18.]

So I ran a proper math test. Had to prove my 2nd research question properly, to see if the ruined mud hills sit systematically closer to roads than the untouched safe hills instead of jst looking at the map image n guessing like a layman.

So I pulled the road files. Used the `osmnx` Python package to download the full drivable road network map for this Kashmir zone (which gave me exactly 44,622 road line pieces), wrote a loop to measure the shortest distance from all 201 hill shapes to the closest road edge, nd compared the spacing lists between broken and intact hills using a one sided Mann Whitney U test because standard t tests break down on this type of non normal data tht has tons of zeros.

The math gave me a very clear proof. My data shows tht the ruined hills sit at an average distance of jst 75.6 meters from a road (nd the median is literally 0.0 meters, meaning more than half of these broken hills are directly touching or cut through by a road!), while the safe hills sit much further back at an average of 133.1 meters with a median of 38.5 meters. My Mann Whitney U test confirmed tht this distance gap is 100% solid nd statistically significant with a low p value of 0.0116.

My 2nd research question is now answered with solid math support. I found tht land destruction is heavily tied to how close a hill sits to the road grid, which makes total sense if u think abt trucks needing easy asphalt access to load n steal the soil. This completely wraps up all the core coding n math scope I planned for this project! (Both of my big starting data gaps, the missing multi year middle trend points n the formal spatial statistics test, are now completely sorted out n closed).

My final output file `data/processed/karewa_road_proximity.gpkg` now holds this whole updated data table with all the distance metrics saved inside. Now, I am officially done with the heavy data analysis part n can move straight into my final deliverables phase where I build the visual charts, export the GIS maps, style the interactive web dashboard, n draft the final research paper paragraphs.

Entry 10

> [Withdrawn in v2. The result described in this entry depended on the withdrawn first-version layer; see the note at the top and Entry 18.]

I ran a massive quality double check. So I chose to review every single script file n web dashboard page inside my Stolen Strata folder to calculate every headline statistic again from scratch using the raw data files.

Then, I caught a huge code break. My main script file `src/analysis/01_extract_karewa_terraces.py` sitting inside the code repository had hardcoded numbers saying `tpi_threshold = 5` and `slope_threshold = 5` (which was the old, rejected filter pair tht only gave 115 useless skinny line shapes!). Every single result I wrote down in my final report (like the 201 proper hills, the 3,305.3 hectares total size, n all

the matching distance tables) came from the relaxed settings where TPI is greater than 3 n slope is under 8 degrees. So, if any external student cloned my GitHub page n tried to run the full code from scratch, they would hve silently gotten tht bad 115-hill output instead of my report numbers! I double checked this error by manually running the DEM to TPI code block again using the raw `StolenStrata_DEM_GLO30.tif` raster image with both setups. The old (5, 5) settings gave exactly 115 pieces, while the proper (3, 8) settings gave exactly 201 shapes (matching my saved geopackage files perfectly). I fixed this mess by typing the correct numbers into the Python script file and added a quick code comment explaining this calibration history so the values can nvr drift away secretly again.

Re calculated every single headline number. Instead of jst loading my old saved geopackage files, I opened the raw image files nd the original OpenStreetMap road data to compute the math statistics all over again from zero.

Everything matched up perfectly. So I independently reproduced the exact four point bare ground trend scores, which came out to 1.84% for 1994, 2.62% for 2005, 2.63% for 2015, n 8.43% for 2025. Then, I verified the total mapped hill area of 3,305.3 hectares. My code also confirmed the total land loss of 190.3 hectares (which is 5.8% of the full map) n verified tht 128.2 hectares of that damage, a massive 67%, sits inside jst those 25 bad hills tht cover a tiny 9.7% of the total space. (Also re ran the raw Saffron Index maps, which confirmed my 14 saffron hills covering 225.4 hectares exactly).

Nxt, I fixed my dashboard gaps. So I re calculated all six points of the proximity risk sensitivity curve: 21%, 29%, 43%, 71%, 86%, n 93% for the 500-meter to 2,500-meter distance buffers. Before this final audit, I only labeled the 500-meter nd 2,500-meter endpoints as confirmed on my dashboard page, while the middle four points were just rough guesses, but now all six are 100% verified n exact. Lastly, I checked my road distance Mann Whitney U test score against the full network of 44,622 road lines, and it gave the exact same 75.6 meters versus 133.1 meters gap with a p value of 0.0116. Found zero math errors or code bugs in this whole batch, which is a much cleaner and happier result than my other portfolio projects where I caught annoying aggregation bugs!

I finally finished my geology math. Before this, my script `src/analysis/10_geomorphometrics_and_figures.py` was already done, but I kept the output hidden behind a stupid `GEOMORPHOMETRICS_CONFIRMED = False` toggle code on the web dashboard while the page just txt printed tht the shape checks were "being finalized." I went back to the raw DEM file n recalculated the compactness shape formula (4 * pi * Area / Perimeter squared) along with the average inside slope angle by myself, nd my new math matched the old charts exactly.

The results gave me a cool new shape proof. My calculations show tht the ruined hills are way less round and compact than the safe, untouched hills (the scores are 0.138 versus 0.191 with a low Mann Whitney p value of 0.0044). This is a completely separate shape based clue tht does not care about satellite greenness or NDVI because it proves these 25 bad hills hve highly jagged, broken boundaries which looks exactly like ugly digging scars from excavator trucks! (But the inside slope angle test showed almost zero difference between the groups, coming out at 2.89 degrees versus 3.06 degrees with a high p value of 0.1711). I am updating this as a fully verified extra result inside `SS_Research_Paper.md` section 3.7 n 4.6 nd across my dashboard pages instead of keeping it as an incomplete task.

I finished my full Stolen Strata double check. I found exactly one bad reproducibility bug inside my scripts nd fixed it immediately (it was jst a wrong setting mismatch between my code n my final map results, so every single headline number I previously wrote down inside my paper was always 100% correct). No other calculation or statistical mismatch turned up in this specific audit round, though this covers only wht I re ran

here, not every possible corner of the project. (Plus, I successfully completed tht extra shape analysis on geomorphometrics tht was hanging left undone before nd pasted it straight into my final website code folders).

Entry 11

> [Withdrawn in v2. The result described in this entry depended on the withdrawn first-version layer; see the note at the top and Entry 18.]

So I went through my work in four separate check rounds, each one looking at a different thing (the heavy math rules, the methodology logic, the file names, n matching it with my other portfolio code). Funnily enough, all four rounds pointed directly at the exact same set of problems, which gave me a very clear hint abt where my true weak spots were hiding instead of throwing four totally random lists at me. (So inside this entry, I am writing down wht I fixed, wht I deeply investigated nd found to be completely correct already, n wht things I chose to skip and leave for future work with a proper logical reason).

All four rounds caught my sloppy choice of cutoff numbers. Initially jst eyeballed the TPI/slope values, the 15-percentage point land loss mark, nd the 0.15 saffron cutoff without running any sensitivity math checks. So I sat down nd coded a brand new script named `src/analysis/11_threshold_sensitivity.py` to test how changing these cutoffs alters my final results.

The data stayed very stable under the hood. When I shifted the degradation limit from 5 to 30 percentage points, my core count of 25 broken hills stayed beautifully steady between 23 and 31 right around the 12-to-20 percentage point neighbor zone. Nxt, I tested the saffron index limit across a 0.05-to-0.25 spectrum. (My 43% danger zone metric barely moved, staying right between 39% nd 44% for almost all values, only going crazy when my sample size dropped below 6 found hills!). Lastly, I checked a big 3x3 grid combining different TPI n slope settings, n the numbers climbed very smoothly from 144 to 279 hills with my selected (3,8) pair sitting safely right in the sweet middle spot. I updated everything inside my papers `SS_Research_Paper.md` section 3.9 nd 4.7.

Stopped using vague excuses. Previously, my draft paper just casually stated tht the sharper 2025 Sentinel-2 pixel size "might slightly increase" my newest trend numbers compared to the old 30-meter Landsat grids from 1994, 2005, n 2015, which was just lazy guessing. I coded a new verification script called `src/analysis/12_robustness_and_effect_sizes.py` to downsample my sharp 2025 images into standard 30-meter blocks using area average calculations nd ran the whole bare mud script all over again.

The numbers gave me the true answer. My calculated average bare earth share dropped slightly from 8.43% down to 7.48%, my total destroyed land area went from 190.3 down to 165.2 hectares, and my count of ruined hills went down from 25 to 23. (So now I know for a fact tht the camera resolution makes a clear 13% difference in the math!). But my main point abt the sudden mining boom still survives this pixel change. Even when I limit everything to the same 30-meter grid size, tht 7.48% score is still roughly 2.8 times higher than the flat, sleepy levels I found during 2005 n 2015. So I immediately went back nd updated `SS_Research_Paper.md` section 4.8, my Limitations paragraph, n the paper Abstract text with these solid math figures instead of using that old, weak, defensive txt.

So I fixed my stats reporting. Previously, I ran three separate Mann Whitney tests checking road distance, shape compactness, nd hill slopes against my degradation tags, but I only wrote down the p values like an amateur. So I used tht same script 12 to calculate rank biserial correlation values for each check (road proximity r is 0.268, compactness r is 0.352, and slope r is -0.170, small to medium, nothing crazy). (So I also

applied a proper Holm Bonferroni correction rule across this 3-test set to make sure my p values were not lucky flukes, and guess what? Both road proximity n compactness survived the cut perfectly, while slope was already useless anyway). So I put these new scores into `SS_Research_Paper.md` section 4.9.

Then I found a very funny writing mistake. In the check rounds I noted tht measuring straight line road distance from the middle center point of a hill is bad n inaccurate for mountain areas, saying I should use the hill's outer edge instead. So I opened `src/analysis/09_road_proximity.py` to check my code against wht I wrote in my draft paper, n I realized my code was perfectly right but my text description was completely wrong! My script was always running `polygon.distance(roads_union)` which calculates the true minimum distance from the polygon boundary edge, nvr the center point. So I went back and corrected `SS_Research_Paper.md` section 3.6 nd the web dashboard's Governance page because both were falsely claiming it was centroid based.

I changed my argument here. Every one of my four check rounds came back to my massive 93% saffron data gap against the official FAO numbers, nd the worry tht this missing data ruins my 43%-within-1km proximity danger finding that I calculated using jst those 14 hills. I added a solid logical argument explaining that a wider, less strict plant picking code would obviously catch much smaller, blurry saffron fields too; nd since those tiny fields are smaller, they probably sit much closer to the messy edge of active dirt quarries, meaning my 43% score is more like a safe, rock bottom lowest floor estimation than an inflated number. (This fits my sensitivity test, where the 43% score stayed steady across different threshold values.) I updated this argument inside `SS_Research_Paper.md` section 5 Discussion n on the dashboard Saffron Vulnerability page.

Leaving my 4th research question fully open felt like a missed chance. The obvious idea was a district table with National Saffron Mission budget targets right nxt to my land destruction map, but I had no real numbers for it. I went online and did a solid search to find proper site level or district wise budget files for this government scheme. (So I found absolutely nothing new besides those same old total valley wide sums I already wrote down, which show 2,598 hectares under rejuvenation and a total cost of 400 crore rupees according to a 2026 Press Post article). There is literally no map ready policy dataset available anywhere tht my hill level map could link up with. Without that data I cannot build the table, so my RQ4 remains an open question. Hve now written down my search attempts directly in the txt so it is clear tht the question is blocked by missing data. So I updated this note inside `SS_Research_Paper.md` section 6 Limitations n on my dashboard Governance page.

Wrote down a big gap in my accuracy check. The check rounds made it clear tht I hve zero formal accuracy testing here: a confusion matrix, producer's accuracy, or user's accuracy calculated against a totally separate set of hand marked ground truth points. (I only did a basic visual check inside QGIS against a few popular known locations). I had never stated this so bluntly before in my draft. Building a proper confusion matrix is a massive task because it requires me to manually look at old, high resolution historical photos n click hundreds of random validation points, which is a full new data collection job, instead of a simple script I can jst run on my current folder files. So I wrote it down as my number one priority for future work. Pasted this txt into `SS_Research_Paper.md` section 6 Limitations.

I forgot a major date warning. My 2005 bare mud percentage score is not a clean, single year photo because my 1st Landsat 5 search gave me zero results, which forced me to stretch the window out to a wide nine year block from 2001 to 2009 centered right on 2005 (Wrote this down earlier in my code diary notes). So I foolishly forgot to mention this as a serious limitation in my main txt, even though it heavily changes how people read my claim abt the land staying completely stable and flat between 2005 and 2015! I typed this out very clearly inside my `SS_Research_Paper.md` Limitations section.

Then, I added a serious warning about my road data. My statistical finding showing tht ruined hills sit much closer to roads is jst a simple correlation test. Hve absolutely zero dated records for road construction to prove if the asphalt came 1st or if the soil miners arrived 1st. (Both situations match up fine with a basic truck accessibility logic). So I wrote this down directly inside `SS_Research_Paper.md` section 5 Discussion nd updated my web dashboard Governance page so the text does not claim cause nd effect.

I skipped a few things on purpose. So I did not code a machine learning model like a Random Forest or XGBoost saffron classifier to fix my low detection numbers because training a whole new model is way too big for a quick review round (n it deserves its own separate future project entry). So I also chose not to calculate a fancy slope weighted least cost path road distance metric to replace my basic straight lines. My current straight line polygon edge distance math is already completely solid on paper nd I wrote it down clearly. (Nxt, I skipped making a hand marked ground truth check because it needs fresh field data collection, and not jst re running old code scripts, though I sorted this out in a later entry!). Finally, I threw out plans for a live online policy calculator slider widget nd basic comparison phrases like "X hectares equals N football fields" because tht is just shiny presentation polish instead of a math or correctness fix. I am saving tht for a lazy day when I want to clean my dashboard look.

So I finished all four rounds, ran code calculations to fix every single serious math issue they caught, like the cutoff sensitivity, pixel resolution differences, statistical effect scores, and that road to hill distance text. The single point I could not fix was tht tricky RQ4 government scheme table. I left it wide open. I wrote down my exact internet search steps inside the report to prove I went looking for district files first, before ever thinking about faking numbers out of thin air just to make the chart look nice and complete. Nxt, I wrote two brand new Python scripts to keep everything organized. They are saved in `src/analysis/`. So I named these files `11_threshold_sensitivity.py` nd `12_robustness_and_effect_sizes.py` so tht my overall project code folder follows the exact same script-1st numbering system I set up at the very start.

Entry 12

> [Withdrawn in v2. The result described in this entry depended on the withdrawn first-version layer; see the note at the top and Entry 18.]

Changed my mind about three items. So I previously listed them as things to do later in my future work section, but they were way too important to leave hanging on a sleepy checklist forever, so I sat down n did a full fresh pass to handle them right away. (These three tasks were a 2nd infrastructure proximity test, a better way to explain my saffron loss numbers to government officers, nd an accuracy check).

Inside this specific diary entry, I am covering town settlement distance, the cash valuation losses, n the current legal status of Karewa hills under local laws. My ground truth accuracy table is tracked in a completely separate file because it still needs me to finish clicking n marking a manual reference sample by hand.

I added a 2nd infrastructure test. My older road check left a big gap because I did not know if soil theft happens only along transport lines or near all human activity generally. To figure this out, I went back to OpenStreetMap n downloaded exactly 3,266 local building outlines for this Kashmir area, then wrote a fresh calculation code script named `src/analysis/14b_settlement_proximity.py` to copy my boundary edge distance method from the road script.

The output was incredibly clear. So I calculated tht the broken hills sit at a mean distance of 455.9 meters from the closest house structure (with a median of 202.0 meters) while the safe hills sit much farther away at a mean distance of 999.7 meters n a median of 816.6 meters. (My Mann Whitney test gave a tiny p value of

0.0001, and the rank biserial correlation value came out to a massive r = 0.465!). This is the single biggest nd most powerful link in my entire research project, beating my road score of r = 0.268 by a mile. I shoved these new metrics straight into my master file `karewa_final_with_geomorphometrics.gpkg` nd added them to my Holm Bonferroni correction script `12_robustness_and_effect_sizes.py` so it now checks a group of four tests instead of three. After running the correction math, house proximity, shape compactness, n road distance all passed the strict test, while hill slope stayed useless. So I pasted these results into `SS_Research_Paper.md` section 3.6, 4.5, nd 4.9, nd generated a brand new Figure 13 chart.

Gave a cash value to my numbers. Before this, my 43%-within-1km danger metric in Section 4.4 only talked about land size in hectares, which does not mean much to politicians. To change this, I searched for the newest 2024-25 J&K state assembly records (the Agriculture Production Department gave an official reply to MLA Hasnain Masoodi stating tht the state made 19.58 metric tons of saffron from 3,715 hectares, with a total value of 534.53 crore rupees). So I ran a quick math check by dividing the harvest weight by the land area (19.58 MT divided by 3,715 ha equals exactly 5.27 kilograms per hectare, which perfectly matches their reported yield number!). Then, instead of picking some random price list from an online shop, I divided the total money by the weight to get a true internal price of 2.73 lakh rupees per kilo, nd wrote everything down inside `src/analysis/15_economic_valuation.py`.

Applied these exact cash numbers to my own mapped farms. My 225.4 hectares of mapped saffron fields make roughly 32.4 crore rupees every year. (Out of tht, the 123.6 hectares sitting inside my 1-kilometer danger radius hold about 17.8 crore rupees of annual crop value, which is a massive 55% of my total!). I wrote this down very carefully as a "value at risk" figure instead of claiming losses tht already happened, because saying the money is already gone would completely break my previous finding that zero saffron fields touch a mine yet. I pasted these cash statistics into `SS_Research_Paper.md` sections 3.5, 4.4, 5, n 6.

Investigated the law books nxt. My 4th research question abt whether the National Saffron Mission investment is target aligned remains a wide open gap because there is no data to answer it. What I did instead was find I could answer a closely related policy question: is there any legal protection covering these Karewa mud hills under current J&K state laws?

My background search hit a total legal void. Right now, there is absolutely zero statutory law protecting Karewas from heavy earth movers, so the whole landscape is a complete free for all. Found tht a private member's bill (introduced by Dr. Syed Bashir Veeri, the MLA for Bijbehara) proposed setting up a strict Karewa Protection Authority, making environmental impact tests mandatory before giving any mining lease, nd slapping major penalties like 10 lakh rupees or 5 years in jail for illegal cutting. But according to the most recent assembly records I could find, that bill remains completely pending, while the Revenue Department nd the Geology & Mining Department jst keep merrily stamping nd handing out the exact excavation permissions that the bill wants to block! (This completely changes how my road distance n house proximity metrics should be read). It proves my maps are showing something else. There's no enforcement breakdown of a broken rule here. What it shows is the stark spatial footprint of a raw extraction industry running wildly inside a completely unregulated open space. I saved these legal notes inside `SS_Research_Paper.md` section 3.8 n 5, and updated the web dashboard Governance & Infrastructure page.

I generated a brand new test dataset during this pass. So I coded a script to select exactly 150 stratified check points, splitting them evenly into 75 marked bare earth spots n 75 marked green crop spots, and saved them at `outputs/ground_truth_sample_points.gpkg` for manual reference checking. (This was the highest priority job from my check rounds, and I am marking them by eye separately since a script cannot automate visual image interpretation). Once I finish clicking nd labeling all these 150 points against a sharp base map, it

will give me the exact confusion matrix, producer's accuracy, n user's accuracy stats that were missing from my draft.

So my coding work is fully complete now. Two new automated scripts named `14b_settlement_proximity.py` and `15_economic_valuation.py` are now added to my pipeline, matching my script-1st numbering system perfectly. So I also updated my older file `12_robustness_and_effect_sizes.py` in its place so it now runs across four statistical tests instead of three because house distance is a core part of my study design now. I went back n edited `SS_Research_Paper.md`, my main `README.md` file, nd all my web dashboard sections so this bigger scope matches everywhere.

Entry 13

> [Withdrawn in v2. The result described in this entry depended on the withdrawn first-version layer; see the note at the top and Entry 18.]

So I completely rebuilt my interactive map layouts. So I wanted every single layer in this project, including the two extra datasets I coded in my last diary entry, to hve both a static picture format and a live web map format, plus I needed a fresh static map for house distance to match the new interactive webpage version.

Threw away the old QGIS2Web software path entirely. That lazy export system was totally broken anyway because it kept messing up my saffron danger zone buffer colors n lines. Now, I wrote a custom script `src/visualization/build_interactive_maps.py` to generate all 8 maps from zero using GeoPandas nd Folium code: covering my study area map, hill destruction status, terrace boundaries, the Lethpora check, saffron danger risk, road network closeness, house settlement closeness, n the saffron cash value at risk.

(I hit a massive file size issue with the road layers because uploading 44,622 separate road line pieces made my map file a giant 11 MB mess tht crashed my browser, so I used a dissolve code to merge them into a single MultiLineString which dropped the size down to just 2.6 MB without losing a single road line at this view zoom!). Manually counted every single shape feature against my raw geopackage tables to ensure the numbers matched 100% before saving the final exports.

So I coded two new fixed map pictures. I wrote a fresh visualization code script named `src/visualization/build_static_maps.py` which successfully spits out two clean image files named `outputs/maps/07_settlement_proximity.png` and `outputs/maps/08_economic_value_at_risk.png`. (To ensure they match perfectly, I opened my old `06_road_network_proximity.png` file, copied the exact hex color codes for the dark background canvas, the panels, n the txt legends, n applied them to my new maps so everything looks uniform). So I did a quick quality check on my six old static maps nd all ten dashboard charts against my current database tables n chose to leave them untouched because none of my core terrace boundaries, road networks, or shape compactness metrics changed at all during this pass. They are completely accurate on their own. I only needed to generate new visuals for my two brand new study layers.

This wraps up my visualization milestone cleanly. Two new fully automated Python scripts join my `src/visualization/` folder under the names `build_interactive_maps.py` n `build_static_maps.py`. Now, every single one of my 8 live web maps and 8 print ready static images comes directly from my own source data arrays instead of depending on tht messy manual QGIS export step. Finally, I went back to my main report file `SS_Research_Paper.md`, added two new image blocks for Figure 14 and Figure 15, nd typed a quick note explaining tht every single static map image has a live interactive twin webpage tht they can open nd click on.

Entry 14

Went out nd stood on the ground for once instead of jst staring at satellite pixels. On 3rd September 2026 I went to Lethpora in Pulwama (same exact saffron belt where my algorithm already lined up with the "Saffron Fields, Lethpora" map label back when I was validating my terrace boundaries) nd took 4 photos with a GPS-stamp overlay on the Lethpora karewa plateau, all clustered tightly around 33.97°N, 74.95°E. [Correction, Oct 2026: these points fall inside the plateau's flat interior, outside all 201 delineated polygons (about 0.5 km from the nearest one), not on a delineated terrace. See Entry 15.]

Wht these photos show needs some explaining tho. Saffron only flowers for a tiny 2-3 week window in October-November, so right now in early September the fields jst look like plain, freshly tilled, dormant soil, not green crop. Tht is completely normal for this point in the season, not a sign tht nothing is planted there. Nd I need to be careful not to mix this up with my degradation classifier: tht bare, tilled look in my photos is NOT the same thing as the bare-earth signal my whole pipeline is built on, because my classifier needs a persistent multi year rise in bare ground, not one random seasonal photo taken at the wrong time of year. This little field check does not prove or disprove degradation either way, it's jst me physically standing on the plateau my satellite map surrounds.

This is also not replacing my bigger 150 point ground truth accuracy sample from Entry 12: tht one is still sitting there unlabeled, exact same as before, still my number one priority job. This was jst a smaller, separate check to go stand on the dirt for once. Planning to go back in October 2026 during peak bloom season to get a proper before nd after pair on the exact same coordinates. Added all 4 photos as a brand new Ground Verification page on the dashboard.

Entry 15

Audit corrections, October 2026. A full re-check of the code, data, figures and documents against the processed data found the following, and these were corrected:

- `12_robustness_and_effect_sizes.py` joined the settlement layer on `terrace_candidate`, which is the raster value from script 01 and equals 1 for every polygon. That produced a 201 x 201 cross join (40,401 rows), so a clean re-run gave settlement p = 0.5 and r = 0. The reported figures (p = 0.0001, r = 0.465) came from the earlier on-disk file and are correct; the script now joins by row position after checking the geometries match, and reproduces them. The Holm loop is now a proper step-down, and its summary line is computed rather than hard-coded.
- The same constant `terrace_candidate` made every interactive-map popup show "Terrace # 1". The maps now use a 1-201 row-based `terrace_id`.
- The TPI/slope rule picks up terrace rims, spurs and edges more than broad flat plateau interiors. At Lethpora the polygons ring the cultivated plateau but leave its interior out, and the Entry 14 photos fall in that interior, about 0.5 km from the nearest polygon. This is now stated as a limitation and as the main reason for the gap against the FAO saffron baseline.
- The claim that compactness "identified the same 25 terraces" was wrong: the 25 least-compact terraces include only 2 of the 25 degraded ones. Compactness is a group-level difference, and because the Copernicus DEM comes from 2011-2015 TanDEM-X acquisitions, it cannot show post-2015 mining scars.
- The rising 500-2,500 m sensitivity curve is cumulative, so it can only go up; it is no longer presented as evidence that the 43% is not a coincidence.

- The National Mission on Saffron's 2,598 ha is area already brought under rejuvenation (of 3,665 ha identified), not a target.
- "No legal framework" is replaced by "no karewa-specific law": excavation proceeds under Revenue and Geology & Mining permissions. The bill's name (J&K Karewa Protection Bill, 2025) and penalties (up to Rs 10 lakh per violation, up to 5 years for repeat offences) now match the cited reporting.
- Static maps 01 (study area) and 04 (Lethpora plausibility check) were QGIS exports: map 01's title carried stray quotation marks, and map 04 filled every terrace in the "Detected Saffron Signature" colour and was titled "Validation". Both are now built by `src/visualization/build_static_maps.py` from the project's own layers (saffron-flagged, other and degraded terraces shown separately, with the field-photo points), over a DEM hillshade, or web tiles when `contextily` and internet are available.
- Other corrections: 2005 window (+-4 years, May-October); trend endpoints are 1994 and 2025; resolution matching cuts net conversion by about 13% and the 2015-2025 jump by about 16%; the TPI window is about 455 m on the 26.8 m grid; the elevation filter removed no polygons; the settlement layer is distinct from roads but not statistically independent; spatial dependence and incomplete OSM building coverage are disclosed; the trend charts use straight segments (the spline invented a dip between 2005 and 2015); slope in script 10 now masks DEM nodata (terrace results unchanged).

Entry 16

Applied Geography rejected the paper. Honestly after the Entry 15 audit I was half expecting it, bcz the biggest problem was sitting right there in my own limitations section: my "terraces" were nvr the terraces. They were the rims. So instead of polishing the same paper again, I chose to rebuild the research itself as a v2, phase by phase, n keep the old v1 files untouched as a record. Everything new goes inside a separate `v2_redesign/` folder.

Phase 1 is the delineation, bcz every other number sits on top of it. My old rule was TPI greater than 3 n slope under 8. The trouble is tht TPI only asks "is this pixel higher than its neighbours within about 455 meters". On a wide flat plateau the middle pixels are the same height as their neighbours, so TPI is zero there n the whole interior gets thrown out. Only the edges nd spurs survive. Tht is exactly why my Lethpora photos were landing outside every polygon while I was literally standing on the karewa.

So the new rule asks a different question: is this ground flat, is it sitting above the nearest stream, nd is it surrounded by scarps? In numbers: slope under 4 degrees, height above nearest drainage (HAND) more than 15 meters, elevation under 1950 meters. HAND was needed bcz my 1st attempt used a simple "height above the lowest point within 8 km" n tht failed badly, the valley floor itself rises toward the south so half the plain started looking elevated. HAND follows the actual water flow path down to the nearest stream, so a sloping valley floor stays near zero while a plateau top cut by nalas comes out 30 to 60 meters.

Tht still was not enough. A big alluvial fan is also flat n also sits above the main river (the Rambiara fan in the south got picked up as one giant 45 sq km "karewa", which is plain wrong). The thing tht separates a karewa from a fan is the scarp: a karewa top ends in a steep drop, a fan jst fades into the plain. So for every flat patch the script now measures wht share of its outer ring is steeper than 6 degrees. 0.45 or more I am calling scarp bounded, 0.20 to 0.45 ambiguous, n below 0.20 rejected. The fan scored 0.05 to 0.11 n dropped out cleanly.

One more fix on the way. My DEM stopped exactly at the study box, so any plateau cut by the box edge had no scarp on tht side n scored too low. I re-exported the Copernicus DEM from Earth Engine with roughly a 10 km buffer (`StolenStrata_DEM_GLO30_buffered.tif`) n reran everything on tht. The big plateau near the airport, which had wrongly gone into rejected, came back as scarp bounded.

Results on the buffered DEM: 49 scarp bounded patches (77.4 sq km inside my box), 24 ambiguous (29.2 sq km), 51 rejected (67.5 sq km). Compare tht with v1: 201 polygons nd only 33.1 sq km. All 4 of my Lethpora photo points now fall inside a scarp bounded plateau. About half of the old rim area lies inside the new plateaus, tracing their edges, which is exactly wht rims should do. The uncomfortable one: 14 of my 25 old "degraded" terraces are not within 100 meters of any plateau at all. They were sitting in Srinagar city nd on the valley floor. So a good part of my v1 degradation result was not even on karewa.

I wanted the GSI geology layer (Karewa Group polygons) as an independent check, but the Bhukosh portal jst timed out on me, so tht is still pending. For the 24 ambiguous patches n the 15 biggest rejected ones, the 1st pass review was done mostly from the terrain (hillshade, elevation, scarps) nd for a handful from satellite imagery. Out of the ambiguous ones, 8 came out as karewa (the big ones, like the Pampore plateau extension toward Dussu nd Ladhu), 8 as not karewa (small bits like the apron around Hari Parbat), nd 8 unsure. I am treating this only as a working decision to move forward. It is NOT validation, n 17 of those 39 calls are low confidence.

With tht, the working v2 terrace layer is `karewa_terraces_v2.gpkg`: 55 polygons, 100.2 sq km inside the study box, each with a proper unique terrace_id this time (no more "every terrace is number 1").

A warning I do not want to forget. When the old NDVI rasters are read on the new plateaus, the bare share goes 0.14% (1994), 0.36% (2005), 0.39% (2015), then 4.65% in 2025 (3.98% at 30 m). Tht is a roughly 12 times jump landing exactly where Landsat hands over to Sentinel-2. It might be real. It might also be the sensor, or the fact tht saffron fields are bare in summer anyway. I am not quoting this number anywhere until Phase 2 rebuilds the series from one sensor family.

For the proper accuracy check I now have a stratified sample of 460 random points (150 on kept plateaus, 100 on excluded flat patches, 150 on other valley land below 2000 meters, 60 within 60 meters of a plateau boundary). The labelling file has no stratum n no predicted class in it, so whoever labels cannot see wht the algorithm said. The key sits in a separate file. Nothing is labelled yet.

Still open in Phase 1: the geology layer, the labelling of the 460 points, n calibrating the three thresholds (4 degrees, 15 meters, 0.45) from those labels instead of from my eye. Scripts for all of this are `01_flat_top_delineation.py`, `02_review_map.py` n `03_finalize_and_sample.py` inside `v2_redesign/phase1_delineation/`.

Entry 17

Bhukosh still would not open. Tried again n again, same timeout. So I stopped waiting for the GSI layer nd went looking for any published karewa map I could lean on instead. The answer was already sitting in my own reference list: Dar nd Zeeden 2020 hve a figure showing where the different Karewa Group sediments lie across the whole valley, redrawn after Bhatt 1982. It is open access, so I can use it with credit.

It is only a picture, not a GIS layer, so it had to be georeferenced. The figure has latitude nd longitude ticks on its border, n those ticks turned out to be evenly spaced (about 1419 pixels per degree both ways), so a plain straight line conversion from pixel to coordinate works. As a sanity check the Srinagar n Anantnag dots on the figure land within roughly 1 to 1.5 km of the true towns. Tht is the real accuracy of this map. It is a schematic valley scale drawing, each pixel is about 100 meters, so I am using it as a rough independent check nd nothing finer than tht. (The capture, the colour classification n the overlay are in the script `04_geology_check.py`.)

Then I simply asked: wht colour does the map show under each of my flat patches?

The result made me happy bcz it was not designed by me in any way. The patches with a high scarp share sit 81% on Karewa formations nd 0% on the green "reworked karewa sediments" class. The rejected ones sit only 30% on Karewa formations nd 36% on reworked sediments, with the rest on recent alluvium. So the thing my scarp rule threw out as "fan" is, on a geologist's map from 1982, mostly material washed off the karewas n laid down again. Tht is exactly wht a fan is. The whole box below 2000 meters is only 39% Karewa formation, so 81% is far above chance. Across the 44 patches big enough for the map to resolve, scarp share nd share on Karewa formations move together (Spearman 0.58).

This also let me set one threshold from evidence instead of from my eye. Earlier I had 0.45 as "definitely karewa" nd a messy ambiguous band below it tht needed someone to look at each polygon. The map says: at a scarp share of 0.25 or more, 90% of the larger patches are on Karewa formations; drop to 0.20 nd it falls to 79%. So the cut is now 0.25, n the manual keep or drop decisions are gone. Only 5 small patches tht pass the rule are still removed by hand (aprons at hill feet nd valley floors inside the mountains, 1.9 sq km in total), n they are listed with reasons in `excluded_by_review.csv`. One call from the earlier review turned out wrong: the big 12 sq km patch in the west tht was marked karewa is 82% reworked sediments on the map. The rule drops it now anyway.

The working layer after this: 57 terraces, 90.9 sq km inside the box.

Now the part I do not like but hve to write. Of everything inside my terraces, 80% is Karewa formation on the map (somewhere between 67% nd 89% if the map is nudged by up to 600 meters, which is within its error). Tht is the good direction. The other direction is weak: my terraces cover only 24% of all the Karewa formation area the map shows in my box, n 38% of the part of it tht is flat. So I am NOT mapping "the karewas". I am mapping the scarp bounded tablelands, the classic flat topped ones. Karewa ground tht slopes gently with no scarp (the Tral valley side is the clearest case) or tht lies low near the valley floor is left out. Part of this gap is the map being a broad brush drawing, but part of it is real, n the paper has to say plainly tht the unit of study is scarp bounded karewa tableland nd not the whole Karewa Group outcrop.

Wht is still not done in Phase 1: the slope nd height above drainage cut offs (4 degrees, 15 meters) are still my choices, the schematic map is too coarse to tune them. The 460 blind sample points are drawn but nobody has labelled them. If Bhukosh ever opens I will rerun this same check on the proper GSI polygons. For now I am calling Phase 1 good enough to build on nd moving to Phase 2, the single sensor time series, bcz tht 12 times jump at the Landsat to Sentinel switch is the next thing tht can sink this.

Entry 18

Phase 2. The question was simple: is the big rise in bare ground real, or did it come from switching Landsat to Sentinel-2 for 2025. To answer it I needed every year from one sensor family, so an Earth Engine script was written for it nd I ran it in my account. Six exports: yearly Landsat NDVI (the 90th percentile of the whole year, so a pixel only counts as bare if it never greens up at all), the June to September median (same as v1), the number of clear observations per pixel, then Landsat 7 alone, Landsat 8/9 alone nd Sentinel-2 alone for the recent years. Two of the six failed on the 1st run bcz of a filter bug in the script (a blank filler image had no date on it). The fixed version is saved as `gee_02_summer_counts.js`.

1st result, n it is the one I was afraid of. The v1 headline is gone. Same v1 polygons, same rule (summer NDVI below 0.15), but Landsat all the way through: 1.15% bare in 1994, 1.11% in 2015, 0.09% in 2025. v1 said 8.43% for 2025. Tht number was Sentinel-2 seeing the ground differently, not the ground changing. So the 190 hectares nd everything built on it (the 25 terraces, the rupee figures) cannot stand as they are.

2nd, the summer median itself is a bad measure here. It jumps from under 1% to 16% nd back within a few years around 2000. Land does not do tht. The counts file shows why: before 2009 a pixel got somewhere between 0 nd 8 clear looks in a whole summer bcz of monsoon cloud, nd in 1991, 1992 nd 1995 basically none. Six years (1990, 1991, 1992, 1995, 1996, 1997) hve too few observations to use at all.

3rd, the level depends on who is looking. For 2025 the yearly measure gives 1.6% with harmonised Landsat, 2.3% with raw Landsat 8/9 nd 4.7% with Sentinel-2. So I can never again write "x% of the karewas are bare". Only change measured inside one sensor means anything.

So I asked a stricter question. Which pixels were clearly vegetated in all three of 2013, 2014, 2015 nd clearly bare in all three of 2023, 2024, 2025, using Landsat 8/9 only. Three years on each side so one dry or cloudy year cannot fake it. Answer: 88.9 hectares on my terraces, which is 0.98% of them. On the other flat raised land tht is not karewa (the fans) it is 6 hectares, 0.07%. In the rest of the box 0.25%. Going the other way, bare to green, 0.8 hectares. Landsat 7 on its own finds 64 hectares by 2019 to 2021 nd Landsat 8 finds 62 for the same window, nd Sentinel-2 agrees tht 98% of those pixels are bare now. Three sensors, same answer. This one is real.

But it is not "the karewas". 76 of the 89 hectares sit on ONE terrace (number 5, about 6.8 sq km, so 11% of it is gone), nd another 11 on the terrace next to it. 52 of my 57 terraces show nothing. It started in 2017 nd 2018 nd kept growing till 2023. I looked at the spot on satellite imagery (33.944 N, 74.848 E, between Auwan Pora nd Tumchi Nowpora) nd it is a field of brick kilns, dozens of them, one is even labelled on the map. Which makes sense, karewa clay is wht bricks are made of. I hve not verified this from any document or on the ground yet, it is one look at imagery.

Wht this means for the paper. The story is no longer a slow valley wide loss. It is tht the earlier number was mostly a sensor artefact, n tht the loss which does survive every check is small, recent nd concentrated in one brick kiln cluster. Smaller claim, but I can defend every part of it. Still open: the 0.25 nd 0.35 cut offs are my choice nd need a sensitivity run, low NDVI does not tell excavation from buildings (tht is Phase 3), nd before 2013 the record is too thin to run the same three year test.

Entry 19

Three follow ups on the Phase 2 finding, all same day.

Sensitivity. I did not want the 89 hectares to hang on my choice of 0.25 nd 0.35, so 24 versions were run: bare cut at 0.20, 0.25 or 0.30, different vegetated cuts, nd both "all 3 years" n "at least 2 of 3 years". The hectares move a lot, from 37 to 151, so the paper has to give a range nd not one number. But two things do not move. In every version the terraces convert at 2.6 to 63 times the rate of the other flat raised land. N in every version 80 to 99% of it is on terraces 5 nd 3. So the size is soft, the location nd the contrast are solid.

Year by year on terrace 5 (`kiln_field_onset_map.png`): 1.3 hectares bare in 2013, 21 in 2017, 66 in 2018, 92 in 2025. It is one compact block at the north tip of the terrace. 46 hectares opened in 2017 to 2018 alone.

Then the part I wanted most nd did not get. I wanted to show the ground actually went DOWN, bcz tht is wht "stolen strata" should mean. Problem: every free DEM is older than the kiln field. SRTM is 2000, ALOS is 2006 to 2011, Copernicus is 2011 to 2015, nd the field opens in 2017. I checked them anyway nd, as expected, the converted block looks jst like the rest of the terrace in all of them, nothing had happened yet. The only free height data after 2017 is the GEDI laser on the space station. Ran tht too. It is accurate enough on the flat terrace tops (noise around 0.7 m), but it only shoots along tracks, nd on my converted block there are 9 usable

shots in total, all from 2019, none later. Those 9 show no lowering (minus 0.3 m against minus 0.5 m on the untouched part of the same terrace), but 9 shots prove nothing either way.

So Phase 4 fails its gate test with free data. I am writing tht down plainly: I can say this land went from vegetated to persistently bare, I can say where nd when, I cannot say how deep or how much volume. ICESat-2 tracks might give a few more points nd I may try them, but I am not going to build the paper on a depth number I do not hve.

Entry 20

Since the whole surviving result now sits on one block, I had to know wht tht block actually is. One look at Google imagery is not evidence. So a search for documents was done nd I am noting wht came back.

The best one is official. A J&K Pollution Control Committee report filed before the National Green Tribunal in July 2024, in a case called Koka Brick Kiln, Rangeen versus UT of J&K. Koka Brick Kiln is the exact name tht shows on the map inside my block. The report says the kiln is at Rangeen Kultreh in Chadoora tehsil, tht it was commissioned in 2017 without consent, tht a closure order was issued in September 2018, nd tht there are 20 brick kilns within 1 km of it. My satellite onset for the block is 2017 to 2018. I did not know about this document when I got tht date, so the two are independent n they agree.

Other pieces: a 2023 article by Raja Muzaffar Bhat says Rangeen Kultreh is karewa land with about two dozen kilns nd tht around a thousand almond trees were cut for new ones. His 2021 piece in Greater Kashmir names Wadipora nd Kultreh as karewas damaged by dozens of kilns. An Air Force letter from 2023 complains the kilns around the station grew fast over the last decade.

One thing does not fit nd I am not hiding it. A 2023 news report says the roughly 24 kilns there were built between 2003 nd 2012 n nothing new came in 2013 to 2022. But on Landsat this block is green every single usable year from 1993 to 2016 nd only goes bare in 2017. Either the older kilns are in another part of the village, or the news timeline is off. The official filing agrees with the satellite. Not settled.

Also none of this has been read in the original yet, so before any of it goes in the paper I hve to open each source nd read it myself. List with links is in `KILN_BLOCK_EVIDENCE.md`.

Saffron. I checked how far the block is from the saffron land I mapped in v1. 6.3 km at the nearest, nd only 0.2 hectares of all the converted pixels are within 1 km of saffron. So the saffron story from v1 does not attach to this loss. Wht is being lost here is karewa under orchards n fields in Budgam, to brick kilns. The saffron belt at Pampore shows no such conversion in 2013 to 2025 by this test.

Entry 21

One worry was left from Phase 1. My terraces cover only about a quarter of the Karewa formation the geology map shows in the box, bcz the rule only keeps scarp bounded flat tops. So maybe the real damage is on karewa ground my rule threw out, nd I am looking in the wrong place. Checked it: ran the same vegetated to bare test over the whole box nd split it by the geology map.

Whole box, 358 hectares converted. Karewa formation inside my terraces: 1.36% of wht was green in 2013 to 2015. Karewa formation outside my terraces: 0.19%. Plain valley alluvium: 0.19%. So the karewa my rule left out is behaving exactly like ordinary valley floor. There is no 2nd hot spot hiding in this box. Good, n a bit of a relief.

Then I went through the biggest clusters one by one, bcz I wanted to know wht this test really catches. Number 1 is the kiln block, 85 hectares. Number 2, 37 hectares on a hillside near Awantipora, is the AIIMS campus being built, clear on the imagery. Then six clusters adding to 58 hectares are INSIDE Dal Lake, which is lake weed turning to open water nd nothing to do with land. Three more next to the kiln block, 25 hectares. The thin lines are road works nd the river shifting its channel.

So two lessons. The test works, it finds real things I can name. But "vegetated to bare" is not "mined". It catches a hospital, a lake nd a road as happily as a brick kiln. In the paper every hectare I call kiln loss has to be backed by looking at it, not by the index alone.

The open question now is the box itself. It was drawn around Pampore for saffron. The one real result is at its western side in Budgam, nd the documents say the main kiln belt (Wathoora, Panzan, Nagam side) runs further west, partly outside my box. If the paper is going to be about karewas nd brick kilns, the box probably has to move west. Tht is a real change of study area, so I am deciding it deliberately nd not drifting into it.

Entry 22

Moved the box west. Old one was 74.75 to 75.15 E, drawn around Pampore. New one is 74.55 to 75.15 E nd goes a bit further south too, so it takes in the Budgam karewas. Ran the Earth Engine exports again for the bigger box (one big Landsat file was still running when I did this, it is not needed for wht follows). Same rules as before, I did not touch a single threshold: 180 terraces, 173 sq km, of which about 80 sq km is new ground.

I expected the west to light up with kilns. It did not. Vegetated in 2013 to 2015 nd bare in 2023 to 2025: 107 hectares on terraces in the whole box, n 90 of those are in the old box. The whole new strip adds 17. The rate on terraces in the new strip is 0.22%, still about 3 times the other flat land there, but small.

The reason is obvious once I think about it. The documents say most Budgam kilns came up between the 1970s nd 2012. Land tht was already bare in 2013 cannot "convert" after 2013. My test only sees new loss.

Wht the wider box did give me: a 2nd site. One terrace near 34.02 N, 74.67 E has about 10 hectares of new conversion, nd on imagery it is brick kilns again. So two kiln fields, both on karewa terraces, n nothing comparable anywhere else in 2,160 sq km. Rangeen Kultreh alone is still 82% of all the terrace conversion. I also tried the same test inside the old Landsat 5 nd 7 years (early 1990s to 2005 to 2007) to catch the older kilns, nd it shows nothing special on terraces, but with only three usable early years I do not trust tht either way.

So the shape of the result after widening: I looked across the whole Budgam to Pampore karewa belt, nd new conversion of karewa tableland after 2013 is rare, 8 terraces out of 180, n where it happens it is brick kilns. It is a sharper nd narrower paper than I planned.

Entry 23

Hve to correct Entry 22 the same day I wrote it. I said the west adds little. Tht was my test, not the ground.

I went to look at the 2nd site on imagery to see wht 10 hectares of kiln looks like. It is not 10 hectares. Zoomed out, the whole tableland around Bandagam, Chand Pora, Hardu Bata Pora, Nijlu nd Bonahama is tan coloured kiln ground with the oval kilns sitting in it, several kilometres across. My strict test had seen almost none of it.

Why: the strict test wants a pixel to drop below 0.25. But a Landsat pixel is 30 meters nd kiln ground is not clean bare soil, it has drying yards, tracks, weeds, a strip of field left in between. Such pixels sit at 0.25 to 0.45. In a 20 sq km window over tht belt the share of pixels under 0.35 goes from 11.6% in 2013 to 18.1% in 2025, while the share under 0.25 only goes from 3.5 to 8.7. So the cut at 0.25 was throwing away most of the thing I am studying. It only caught Rangeen Kultreh so well bcz tht block was cleared all at once nd completely.

So a 2nd test, built on the size of the fall instead of a fixed floor: clearly green in 2013 to 2015 (0.45 or more), under 0.40 in 2023 to 2025, n a fall of at least 0.20. With tht: 163 hectares on terraces in the old box nd 166 in the new strip, 329 together. Run the same rule backwards (land going from low to green) nd it gives 11 nd 31, so about 286 net. 27 terraces out of 180 lose a hectare or more, 13 lose 5 or more, nd a whole bunch of them sit together in the Bandagam belt. Landsat 7 agrees on 86% of the pixels. In the new strip most of the fall comes after 2021.

This test is looser so it is also noisier. Outside the terraces it throws up a lot of change in both directions, which is fields nd building nd wht not. On terraces the reverse change is small, which is why I trust it there more, but I cannot jst swap one number for the other.

Where tht leaves me: two numbers. 107 hectares is a floor, land tht went completely bare, almost all one block. Around 290 to 330 is the wider estimate, two kiln belts. Before either goes in the paper I need to check a sample of the flagged pixels by eye against imagery nd report how many are really kiln ground. Lesson for me: I trusted a threshold bcz it gave a clean answer in the 1st place I tried it.

Entry 24

Accuracy check, done by hand, by me. 120 random points on the terraces: 25 from land the strict test flagged, 60 from land only the looser drop test flagged, 35 from land neither flagged. The points file did not say which group a point came from, tht was kept in a separate key file I did not open. For each point I opened the 2026 satellite image nd wrote wht is at the centre.

1st attempt in the in-app browser failed (the map would not draw unless the window was in front), so I did it myself in Chrome from a small labelling page.

It took two passes, nd the 1st one taught me something about my own labelling. I had seven classes. After the 1st dozen points I almost stopped using "kiln ground" nd was calling everything brown "other bare ground", even points sitting 200 meters from ones I had called kiln a few minutes before, inside the same block. At tht zoom, if there is no kiln or brick row right at the centre, worked kiln ground nd any other bare ground look the same. So the kiln versus bare split from pass 1 was useless. Pass 2: only the 68 points I had called bare or road, map opened one step zoomed out, one question, is this point inside a kiln field or not. I got confused on the edge cases nd took a second opinion on roughly eight points across both passes (noted in the folder), the rest I did alone.

Result. Strict test, 25 points: all 25 are not vegetated today. 19 kiln, 4 other bare, 2 road. Drop only, 60 points: 51 not vegetated, 9 still green. Of the 51, 16 kiln, 17 buildings or roads, 16 other bare, 2 unclear. Not flagged, 35 points: 22 green, 1 kiln, the rest bare or built.

So both tests are right tht the green is gone (100% nd 85%). [Note, 9 Oct 2026: the 85% counts the two unclear drop-only points as not vegetated. Counting only clear labels it is 49 of 60, 82%, which is the figure in the paper.] Where they differ is wht it became. The strict test is mostly kilns, three out of four. The loose test is a mix, only about one in four is a kiln, as much again is houses nd roads.

Putting it together: of the land my tests flag on terraces, about 141 hectares is kiln ground, somewhere between 110 nd 172. Tht is the number I can stand behind now. It is higher than the 107 floor nd much lower than the 329 gross, which is about wht I should hve expected.

Wht this check can not do. It shows wht is there today, not tht it was green in 2013. One person labelled it. Over a third of the unflagged points also look bare in the image, bcz a dry season picture of a karewa is brown anyway. N for kiln ground tht my tests missed I hve one point out of 35, which gives a range so wide (86 to 2,470 hectares) tht it says nothing.

Entry 25

Went through the list of weak points tht could still be fixed nd did the ones tht need only computing.

The two cut offs I chose by hand for wht counts as a terrace (flatter than 4 degrees, more than 15 meters above the drainage). Tried 3, 4, 5 degrees against 10, 15, 20 meters, nine versions. The terrace area changes a lot, 106 to 258 sq km. The result does not: strict conversion stays between 95 nd 120 hectares, nd terraces always convert 5 to 13 times faster than other flat land. Height above drainage barely matters, slope matters more. Same exercise for the three numbers inside the loose test, 27 versions, net loss between 234 nd 374 hectares. So none of my hand picked numbers is carrying the result.

Then the thing I had been missing. The last big Landsat file finally came through, so I could put 1993 to 2025 together as one sensor family for the wide box. This shows the older kilns my 2013 onwards test could never see. On the Bandagam to Batapora terraces, land tht never greens up properly was about 100 hectares in the mid 1990s, 300 by 2008 to 2012, 378 now. It grew in two steps. Rangeen Kultreh is different, basically zero until 2015 nd then 100 plus hectares in three years. Over the whole thirty years the two belts together lost about 335 hectares of green terrace land. Meanwhile all the other terraces, 140 sq km of them, did not lose anything, they come out a little greener.

One caution on tht last line. Everything looks a bit greener from 2013 on bcz Landsat 8 gives more looks per year, so the "greener elsewhere" part is partly the instrument. It works against my finding, not for it, so the 335 is on the safe side.

Looked for documents on the Bandagam belt nd found none tht name those villages. Only district numbers: the Pollution Control Committee counted 213 kilns in Budgam in April 2023 nd said most outside the airport radius had no valid consent. So for tht belt I hve imagery nd the district count, no village level paper.

Not done: the geology check for the far west strip, the browser would not cooperate. Still needs a human: checking old imagery to confirm the flagged points were green before 2015 (the files for it are ready), nd a 2nd person labelling the 120 points.

Entry 26

Rewrote the paper nd the executive summary today, start to finish. Not an edit, a new paper under the old name.

The hardest line to write was the 1st one after my name: tht the numbers in the September preprint are withdrawn. 1.84 to 8.43 percent, the 190 hectares, the 25 degraded terraces, the 17.8 crore of saffron at risk, the road nd settlement tests. All of it stood on one layer nd tht layer was a satellite switch. I put the correction up front nd gave it its own results section with the figure, bcz anyone who read the preprint should see exactly how it fell, not jst tht it did.

Wht the paper says now. 180 scarp bounded tablelands, 173 sq km, mapped by slope, height above drainage nd scarp share, checked against the Dar nd Zeeden geology map. Yearly peak NDVI for every pixel from 1993 to 2025, Landsat only. Two tests for land tht was green in 2013 to 2015 nd is not in 2023 to 2025: 107 hectares strict, 329 loose. My 120 hand labelled points say about 141 hectares of tht is brick kiln ground, give or take thirty. Two places carry it, Rangeen Kultreh, which did not exist as a kiln field before 2016, nd the Bandagam to Batapora tablelands, which hve been going since the 1990s. Thirty years, about 335 hectares between them. The other 140 sq km of tablelands, nothing.

The title changed too. "Threat to the saffron economy" is gone from it bcz I could not find tht threat in the data. The saffron tablelands at Pampore are not being converted this way, at least not since 2013. I kept "Stolen Strata" bcz it is still true, only in Budgam nd for bricks.

Things I made myself write down in the limitations even though they hurt the paper: I map a quarter of the karewa formation, not the karewas. I cannot say how deep anything was dug. My accuracy sample shows today nd not 2013. I labelled it alone nd took a second opinion on roughly eight points. Some of the news nd tribunal facts I still hve to read in the original myself before this goes anywhere.

Still to do before it can be submitted or the preprint replaced: read every cited source in the original, the old imagery check on the sample points, a 2nd labeller if I can find one, new PDFs, nd the README nd the EarthArXiv coversheet version, which still carry the old numbers. Figures for the new paper are in `v2_redesign/paper_figures`.

Entry 27

The old imagery check, finally. This was the hole I kept writing into the limitations: my 120 points show wht is there today, not tht the land was green before.

Only the 85 flagged points matter for this (25 strict, 60 loose), the 35 unflagged ones were never claimed to hve changed. I opened each point in Google Earth Pro with the historical imagery slider. This part needs no special skill, where it is green it is vegetated, nd tht is how it went. I kept the slider on September 2014 for almost everything, 77 points. For eight points tht image was under cloud, in a cloud shadow or too hazy, so those are on a clear image between June 2013 nd July 2014 nd the date is written per point.

How the labels were made. I took a screen capture of each point, got a first-pass label on the capture, nd then confirmed or changed it myself on screen. This was not blind, I knew all 85 were flagged. Several points did not get a label on the 1st capture: cloudy ones were held back until I found a clear date, edge points until I looked from directly above, nd points sitting under the grey road overlay until I switched tht layer off (it draws today's roads over the 2014 picture). Two things I learnt the slow way: zooming in too far makes a 2014 image worse not better, nd the date on the slider is not always the date of the picture, the status bar is.

Result. 72 of 85 show vegetation in 2013 to 2014. All 25 strict points. 47 of the 60 loose ones. 12 loose points were not green nd 1 I could not tell.

The 12 are four groups of three. Already kiln ground or brick rows in 2014 (27, 30, 32). Buildings or hard ground in a compound or settlement (79, 87, 114). Bare ground nd tracks next to sheds (8, 39, 69). Bare soil plots (20, 78, 106). Point 78 is small bed plots, proper farmland, jst bare in a June picture.

Wht this does to my number. Two of those twelve (27 nd 30) are points I counted as kiln field today. If I only count kiln points where the old picture also shows green, 141 hectares becomes about 134, somewhere between 103 nd 164. I am keeping both in the paper: 141 is wht the satellite test gives, 134 is wht survives an

independent picture of the before state. Land tht was green then nd is not green now, whatever it became, is about 245 hectares of the 331 flagged.

Wht I should not oversell. "Vegetated" here is a low bar. At Rangeen Kultreh the before picture is scattered trees nd bushes on dry ground, not thick green. Nd one picture is one day, while my test uses the greenest time of the year, so a point bare in one image is not proof the satellite was wrong, nd a green one is not proof it was right for the whole year. The strict test comes out of this very well, 25 out of 25 both ways. The loose test is confirmed as loose: about one in five of its points was not clearly green to begin with.

Files: `accuracy_before_pass.csv` (label, image date nd wht is seen at each point) nd `10_before_check_result.py` in `v2_redesign/west_extension`. Paper section 4.5 has a new Table 4. Still open from the old list: read every cited source in the original, nd a 2nd labeller. [Update: the two official filings were read in full on 7 October, Entry 28.]

Entry 28

7 October 2026. Sources, a check on the removed flat tops, and a second sample.

I read the two official documents in full: the Pollution Control Committee report in the tribunal case of the Koka kiln at Rangeen (OA 364/2024) and the Committee's status report with the district list of kilns (OA 594/2022, 4 October 2023). The 2017 commissioning without consent, the closure order of 10 September 2018 and the 20 kilns within one kilometre are as I had them. The district list has six entries for Kultreh, three of them not yet built, so it does not support two dozen older kilns under that village's name. It does list kilns under most of the Bandagam–Batapora village names (Chandpora, Bonhama, Batapora, Harda Batapora, Bandgam, Nigloo), without locations. The press reports I have checked only against extracts, and the paper says so. Six statements in the paper were corrected against the originals.

The four flat tops removed by review in the wide box cover 0.5 km² inside the box and hold 0.09 ha of strict-test conversion and none by the drop test (`11_removed_flat_tops_check.py`), so no reported figure depends on that review.

I drew a second random sample of 120 points from the same three strata (35 strict, 60 drop-only, 25 unflagged; `12_accuracy_sample_supplement.py`), to be pooled with the first. The stratum key is in a separate file that stays closed until all points are labelled. For this sample I record, for each point, the date of the newest image as well as the 2014 image, and only "inside a kiln field or not" for the present state. By the evening 93 of the 120 were labelled.

Entry 29

9 October 2026. A full review of the whole project, and the fixes that followed.

I went through the whole project again end to end, read-only first, and recomputed Table 1, the sensor cross-checks, the long series and the Section 4.1 check from the raw rasters. All of them matched. The review also turned up things I had missed:

- The headline that terraces convert several times faster than other flat land comes entirely from the two kiln belts. Outside them, and compared polygon by polygon, terraces convert at the same rate as comparable land (`13_polygon_level_comparison.py`). The paper now says that.
- Bandagam–Batapora is not only the older belt. In the Landsat 8/9 record it has the larger post-2013 loss by the drop test (142 ha against 114 ha at Rangeen Kultreh). My own notes from the second sample

point the same way.
- The check that omitted karewa behaves like valley floor had used only the strict test. Rerun with both tests on the wide box as far west as the geological map reaches, the conclusion holds (`14_omitted_karewa_both_tests.py`).
- The Earth Engine scripts apply Roy et al.'s ETM+ → OLI coefficients to OLI data, the reverse direction. Only the long series is affected. It has to be re-exported.
- The "2026 imagery" in the accuracy section was the date I looked, not the date of the image. The paper now says so; the image dates of the first sample still have to be recorded.
- Several numbers had come from the first, smaller box and disagreed with the wide-box values (Rangeen Kultreh 21/66 ha, the 15% of 2001); these were replaced.

The paper now states three research questions, the abstract starts from them, there is a short section on earlier satellite work on brick kilns and on Landsat change detection, and the limitations list the items above. Still to do by me: finish and pool the second sample, record image dates, write down what I counted as "inside a kiln field", label the terrace reference sample, re-export the long series, and a second labeller.

Entry 30

9 October 2026. The second sample is finished and pooled with the first.

All 120 points of the second sample are labelled, each from two screen captures (the newest image and the September 2014 image, or the nearest clear image of 2013–2015 where September was cloudy), with both dates recorded. Only then did I open the stratum key. Two points had no clear image in 2013–2015 (173 and 181); they count as unclear for the earlier state. Both are unflagged points, so the before-check of the flagged land is not affected. The pooled estimate is in `15_pooled_accuracy.py`.

Pooled, the strata hold 60, 120 and 60 points. The loss of vegetation holds up well: 299 ha of the 331 ha flagged is not vegetated today (interval about 285 to 313 ha), and 260 ha was vegetated in the older image and is not vegetated now (241 to 279 ha). The two samples agree on these to within about 30 ha.

The kiln figure does not hold up as well. Pooled it is 188 ha (166 to 210 ha), but the first sample alone gives 141 ha and the second 232 ha, and in the drop-only stratum the difference is too large to be chance (16 of 60 points against 35 of 60). It is not that the points are different: kiln, other bare ground and road together take 40 points in the first sample and 43 in the second. The difference is where I put bare worked ground, tracks and cleared plots at the edge of a kiln field. In the first sample most of them went to "other bare" or "road"; in the second, judged in one pass against the setting, most went to "kiln". So the kiln share depends on where the edge of a kiln field is drawn. The paper now gives 188 ha with both sample figures beside it, and puts the loss of vegetation first. Writing down a rule for the edge of a kiln field, or drawing the kiln fields on dated imagery, is the next step for this number.

Changed: paper sections 3.5, 4.5 (Tables 2 to 4), 6 and 7 and the abstract; executive summary; README; dashboard. The first sample's own tables and scripts (05, 09, 10) are kept as they were, as the record of that sample.

Entry 31

10 October 2026. One rule for where a kiln field ends, and the kiln figure settles.

Pooling the two samples had shown that they put the edges of kiln fields in different places, so the kiln figure was 141 ha by one and 232 ha by the other. I wrote the rule down (`KILN_FIELD_RULE.md`): a point is kiln if it is on a kiln, a pit, brick rows or a kiln shed, or on worked ground, a track or a yard inside the worked area with bricks, a pit or a kiln within about 100 m and nothing like a house, a public road or a crop field in between. Then I went back to every flagged point whose label had depended on its setting, 57 from the first sample and 29 from the second, and labelled each again under the rule from a view straight down on the newest image.

27 labels changed. Sixteen first-sample points moved into a kiln field: tracks and worked plots that I had called bare or road the first time. Four second-sample points moved out: three cleared or cut plots with nothing kiln-like within 100 m (148, 213, 238) and one too dark to tell (159). Point 40 I first read as bare ground on a wide view and then, closer, found brick stacks about 60 m away.

With the same rule the two samples agree: 204 ha and 217 ha, 210 ha pooled (189 to 231 ha), and 197 ha on land the older image shows as vegetated. The loss of vegetation does not move (299 ha). This pass was not blind to stratum, and the rule was written after I had seen the two samples disagree, which the paper says. Most of the kiln land is still recognised from its setting within 100 m of bricks or a kiln, not from what lies at the point; outlines of the kiln fields on dated imagery would be the independent check.

Entry 32

10 October 2026. The long series, recomputed the right way round.

On 9 October I had found that the long series turned Landsat 8/9 into Landsat 7 with the wrong set of Roy et al. coefficients, the ones meant for going from Landsat 7 to Landsat 8. I wrote a separate script for the fix (`gee_05_long_series_fixed.js`) instead of editing the old one, so the first run can still be reproduced. It exports only the 2008–2025 stack, since Landsat 8 starts in 2013, with the OLI to ETM+ coefficients (red 0.9372 x + 0.0123, near-infrared 0.8339 x + 0.0448). Then I reran `08_long_series_wide.py`, `build_data.py` and `make_figures.py`.

Nothing from 2013 to 2025 moves, because the conversion tests use Landsat 8/9 alone. The long view does move. The step in peak NDVI at 2013 is smaller (the terrace median goes from 0.58 to 0.64, not to 0.67), so less of the late period reads as greening. Net of change in the other direction, the drop test from 1993–1998 to 2023–2025 now gives +98 ha at Rangeen Kultreh and +309 ha at Bandagam–Batapora, about 407 ha together instead of 335, and −202 ha on the other terraces instead of −336. Bandagam–Batapora's low-NDVI land is now 454 ha in 2023–2025, not 378. I still treat the three-decade figure as indicative, for the same reasons as before.